

Haetham AL ASWAD

Post-doctorant in cryptography

Personal Information

- Email : haetham.al-aswad@lirmm.fr
- Webpage : hal-aswad.github.io

Post-doctoral

Feb 2025 - **Class Group Computation In Quadratic Fields**, *CNRS Montpellier, France*
Ongoing I work with Cyrille Bouvier, Laurent Imbert and Pascal Giorgi on computing smith normal form for sparse matrices. I also work with Katharina Boudgoust on the Module-LIP problem.

PhD Thesis

Oct 2021 - **The Discrete Logarithm Problem in Extensions of Finite Fields**, *INRIA Nancy, France*
Dec 2024 Under the supervision of Emmanuel Thomé and Cécile Pierrot. I focus on studying the Discrete Logarithm problem, specifically in non-prime finite fields (with extension degree greater than one). My thesis contributed to improving the Number Field Sieve algorithm and its variants in both theoretical and practical aspects.

Education

2020 - 2021 **Master 2 in Applied algebra**, *University of Versailles, France*
2020 **Degree of agrégation in Mathematics**, *France*
2019 - 2020 **Master 2 preparation of the agrégation in Mathematics**, *University of Saclay, France*
2018 - 2019 **Master 1 in fundamental Mathematics**, *University of Saclay, France*
2017 - 2018 **Bachelor's degree in fundamental and applied Mathematics**, *University of Saclay, France*

Publications and submitted work

Al Aswad, Pierrot, Thomé **Discrete Logarithm Factory**, *IACR Communications in Cryptology*, 2024
Al Aswad, Pierrot **Individual Discrete Logarithm with Sublattice Reduction**, *Designs, Codes and Cryptography*, 2023

[Submitted work](#)

Al Aswad, **Accelerating the Tower Number Field Sieve with Galois automorphisms**,
Pierrot, 2025
Thomé Available at https://hal-aswad.github.io/assets/accelerating_tnfs_with_galois_automorphisms.pdf

Invited talks

Slides made with the animation library Manim and available on my webpage.

Oct 2025 **Almasty Team, Paris, France**

Using Galois automorphisms to accelerate TNFS.

Feb 2025 **Eco Team, LIRMM, Montpellier, France**

Using Galois automorphisms to accelerate TNFS.

Oct 2024 **Géométrie et algèbre effective Team, IRMAR, Rennes, France**

Using Galois automorphisms to accelerate TNFS.

Internships, Projects, and Fellowships

May 2022 - **Research stay at the University of California as part of the thesis, UCSD,**
June 2022 San Diego

under the supervision of Nadia Heninger and Emmanuel Thomé. I Worked on my Phd and discussed lattice related problems with Nadia Heninger and Phd's students Adam Suhl and Keegan Rayan. This tenure was sponsored by the program *Dream* of University of Lorraine

Sep-2022 **Founder of a monthly seminar for PhD students, LORIA, Nancy, France**

The seminar aims to open scientific discussions among PhD students across diverse computer science domains within the LORIA laboratory

Internships

March 2021 **The individual Logarithm step in NFS, INRIA Nancy, France**

- Sept 2021 under the supervision of Cécile Pierrot

2019, 40h **Teaching in high school, lycée des loges Evry, France**

2019 **Study of group representation's theory and an application to random walks on finite groups, University of Saclay, France**

under the supervision of Amaury Freslon

August **Study of the quality of digital hand signatures using fractal dimensions,**

2018 *Institute of Mines Télécom Evry, France*

under the supervision of Nesma Houmani

Summer Schools

Nov 2022 - **REDOCS : Constructing an authentication protocol, CIRM, France**

Dec 2022 under the supervision of Chloé Héban. Development of an authentication protocol designed to address a real-world challenge presented by the company *Cosmian*. This is a co-work with four other Phd-students

August **MathInFoly, Insa Lyon, France**

2019 Introduction to cryptography and proofs with Coq

Implementations

- Nov 2023 **Implementation in Sage of the Tower Number Field Sieve**, *INRIA*, France
Co-project with Cécile Pieroot and Emmanuel Thomé
- Dec 2020 - **Implementation in C of the Hellman-Reyneri algorithm for the discrete**
Feb 2021 **logarithm in small characteristics**, *University of Versailles*, France
Co-project with Hadrien Notarantonio

Fellowships

- Oct-2021 - **INRIA fellowship for PhD**
Sep 2024
- Oct-2021 - **French Ministry of Army fellowship for PhD**
Sep 2024
- May-2022 - **Mobility fellowship Dream**, *University of Lorraine*, France
June 2024 Two months stay at the University of California San Diego

Other

- 2016 **Co-author of the comic book Haytham une jeunesse syrienne**, *édition Dargaud*
Comic book co-written with the writer Nicolas Hénin and the cartoonist Kuyng Park

Teaching

Summer school

- July 2024 **Discrete logarithm in finite fields**, *6h*, École CIMPA, Douala, Cameroun
A course on the discrete logarithm problem for master students in Mathematics

Lectures

- Nov-Dec 2023, June 2024 **Graph theory : shortest path problem (Master 1)**, *18h*, École des mines Nancy, France
Study of three algorithms, Bellman-Ford, Dijkstra, and A*, with a focus on correctness and optimality proofs. The notes are available on my webpage

Exercise sessions

- 2021 -2024 **Exercise sessions in the following courses for three years**, *64h per year*, École des mines Nancy, France
- Programming in Python and Data Structure (Bachelor 3).
 - Advanced Algorithms and Complexities (Bachelor 3).
 - Operations Research (Bachelor 3).
 - Introduction to Machine Learning (Master 1).
 - Programming Languages : JavaScript and Go (Master 1).
- 2019 - 2021 **Mathematics (kholles for Bachelor 1)**, *2h per week*, Blaise Pascal Orsay, France

- 2017 - 2018 **Preparation for Science at University**, *20h*, University of Saclay, France

Supervision of students

- Jan 2022 **Studying primality tests : "Observation Internship" for a ninth grader**, *35h*, INRIA Nancy, France

Languages

Arabic and French (bilingual)

English (fluent)